# MRO Reverse Engineering Study

## Assignment #2

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# Contents

# 1 Mission Analysis

## 1.1 Mission Trajectory Design

In the next section, mission trajectory design is analyzed, dividing the flight into phases and reporting the $\Delta V$ costs for each maneuver.

### 1.1.1 Low Earth Operations and Cruise Phases

During the cruise phase, Mars Reconnaissance Orbiter (MRO) followed a Type-I ballistic trajectory between Earth and Mars, with a change in true anomaly less than 180° between its departure (12/08/2005) and arrival (10/03/2006) dates. The interplanetary trajectory met these constraints [1]:

- 21-day launch window for high success probability.
- Excess velocity under 3 km/s for cost-effective orbital capture.
- Arrival date and nodal orientation suitable for aerobraking into the Primary Science Orbit.

Atlas V was selected for its efficiency with a high Declination of Launch Asymptote (DLA) ($\geq 41°$) and high injection energies (C3 $\geq 20\,\mathrm{km^2/s^2}$)[1]. Five Trajectory Correction Maneuvers (TCMs) were planned based on covariance analyses predicting navigation uncertainties. TCM-1 corrected the upper stage injection errors of the launcher, while TCM-2 addressed the execution errors of TCM-1 and the uncertainties of residual orbit determination. TCM-3 and TCM-4 were planned as drift mitigation. A contingency maneuver (TCM-5) was included to prevent an unexpectedly low orbit altitude [9].

### 1.1.2 Mars Orbit Insertion and Aerobraking

On 10/03/2006, the spacecraft began the Mars Orbit Insertion (MOI) maneuver, designed to match the hyperbolic trajectory from launch and cruise with the starting conditions of aerobraking. The maneuver placed MRO into an high-elliptical orbit with apoapsis at 44 982 km, periapsis at 426 km, and a period of 35.5 h. On 30/03/2006, aerobraking began, aimed at lowering the apoapsis to 486 km. This maneuver exploits the effects of atmospheric drag, allowing to save up to 1200 m/s in $\Delta V$. During the aerobraking various maneuvers were planned [1][9]:

- **Corridor Control Aerobraking Maneuver (ABM)**: Aims at raising and lowering the periapsis. 26 of these were performed, with a total planned $\Delta V = 33\,\mathrm{m/s}$. In particular, the first six of them lowered the periapsis from 426 km to 108 km during the walk-in, the initial stage of aerobraking.
- **Pop-up maneuver**: Used to raise periapsis to 160 km in case of unsafe anomalies, with $\Delta V = 20\,\mathrm{m/s}$;
- **Aerobraking Termination Maneuver (ABX)**: Performed during the walk-out phase with $\Delta V = 25\,\mathrm{m/s}$, raising periapsis from 111 km to 215 km, marking the end of the aerobraking phase.

### 1.1.3 Transition and Primary Science Orbit

After aerobraking, MRO executed a series of Orbit Adjustments (OAs) maneuvers to reach the Primary Science Orbit (PSO). OA-1 raised periapsis and adjusted inclination, while OA-2 lowered apoapsis and rotated the apse-line from an argument of pericenter of 187° to 270° [9]. The PSO was designed to [1]:

- Enable low-altitude observations.
- Maintain a repeating ground track for rapid mapping, short-term revisit opportunities, and long-term track spacing under 5 km at the equator.
- Minimize atmospheric drag effects.

- Be sun-synchronous for repeatable lighting conditions.

The selected orbit had a Sun-synchronous ascending node at 3 P.M. LMST, a near-polar inclination of 92.7°, and an eccentricity that maintained a low-altitude frozen orbit (255 km periapsis, 320 km apoapsis, 270° argument of periapsis) [9]. Its 3775 km semi-major axis ensured a 17-day ground track repeat cycle. In this phase, Orbit Trim Maneuvers (OTMs) are planned to maintain the sun-synchronous orbit and the repeating ground track. In addition, a $\Delta V$ budget equal to 19 m/s has been allocated to station keeping, and the same quantity for the station keeping of the relay phase [1].

### 1.2 Mission Maneuver Sizing

This section examines mission trajectory design, calculating and justifying $\Delta V$ costs through maneuver analysis and station-keeping. Using patched conics method and an impulsive maneuver assumption, maneuvers are assessed with their respective $\Delta V$. Computations of TCM-1 and TCM-2 are implemented based on the propagation of the uncertainty of launcher injection [7] and the uncertainty of TCM-1, respectively [1]. Due to their stochastic nature, 100 % margin has been applied to their costs. For MOI, velocity at the entry of the sphere of influence is determined via a Lambert-based algorithm. The estimated, not margin-adjusted, $\Delta V$ for insertion, computed as a maneuver at pericenter, is lower than the real one, due to non-impulsive corrections. Available data allowed to estimate accurately the first six ABMs. These, aimed at lowering the periapsis to 108 km, were modeled as apoapsis maneuvers. During aerobraking, the saved $\Delta V$ is estimated via energy balance between the initial and final orbits. The estimated result of saved cost ($\Delta V = 1994$ m/s), deviates greatly from the real one ($\Delta V = 1182$ m/s). This discrepancy is due to the rough approximation given by the energy approach, which does not take into account many properties of the aerobraking maneuver. A more precise atmospheric drag model would give a more reliable estimation. OA-1 was modeled as a plane change, while OA-2 as an apse-line rotation. Deviations are mainly inked to environmental perturbations.

### 1.3 $\Delta V$ Budget Breakdown

Margins follow industry standards [6], based on whether the maneuver is stochastic or deterministic.

| **Maneuver** | **Real $\Delta V$** [m/s] | **Estimated $\Delta V$** [m/s] | **Margins** | **Total $\Delta V$** [m/s] |
|---|---|---|---|---|
| **Cruise Phase** | | | | |
| TCM-1 | 7.80 | 15.7 | 100% | 31.4 |
| TCM-2 | 0.754 | 0.129 | 100% | 0.258 |
| **MOI** | 1000.5 | 952.9 | 5% | 1000.5 |
| **ABM** | | | | |
| ABM-1 | - | 4.1 | 5% | 4.3 |
| ABM-2 | - | 4.2 | 5% | 4.4 |
| ABM (3-6) | - | 6.0 | 5% | 6.3 |
| **Total** | 14.5 | 14.3 | 5% | 15.0 |
| **ABX** | 25.0 | 26.2 | 5% | 27.5 |
| **Transition to Science Orbit** | | | | |
| OA-1 | 15.0 | 18.5 | 5% | 19.4 |
| OA-2 | 53.5 | 50.6 | 5% | 53.1 |
| **Station Keeping** | 38.0 | 45.4 | 100% | 90.8 |

Table 1: Mission Maneuver $\Delta V$ Budget

# 2 Propulsion Subsystem

The MRO propulsion system is required to perform orbit maneuvers of varying magnitudes, trajectory correction maneuvers, attitude control, station keeping, and momentum desaturation of the reaction wheels. The primary drivers that significantly influenced the design and architecture of the propulsion system are:

- The mission's reliance on highly critical maneuvers.
- The high frequency of maneuvers over time, particularly during the aerobraking phase.

To meet these requirements, MRO is equipped with a total of 20 monopropellant hydrazine thrusters of three different sizes. The propulsion system operates with blow-down pressurization for all thruster activities except for the MOI burn, where a pressure regulator is employed to ensure consistent thruster performance during this risky maneuver [8].

## 2.1 Architecture and rationale

### 2.1.1 Thrusters

To ensure greater robustness, hence being more tolerant to single-fault thruster failures, a configuration with multiple smaller engines is preferred over a single, more powerful thruster [4]. The need to deliver impulses of different magnitudes and in different directions throughout the mission led to the choice of using three distinct types of thrusters as detailed in Table 2.

- The primary engine thrusters (MR-107N) are specifically designed for the MOI burn where a large amount of $\Delta V$ should be quickly delivered, firing simultaneously all the 6 engines [3]. To mitigate the risk associated with first-time usage during MOI phase, also TCM-1 is executed using the MR-107N [3][4]. These thrusters are aligned along the principal axis of the spacecraft.
- The intermediate thrusters (MR-106E) are mainly designed for the TCMs during the cruise phase but they are also used for orbital parameter adjustments during and after the aerobraking phase, thrust vector control during MOI and Station Keeping maneuvers during the Primary Science Phase (PSP) and Relay phases [3][4]. Due to their dual-role functionality, these thrusters serve both as primary and secondary propulsion. These thrusters are strategically distributed around the spacecraft, oriented to provide thrust in multiple directions, thereby generating both translational and rotational accelerations.
- The smallest thrusters (MR-103D) are primarily utilized for attitude control and for executing Angular Momentum Desaturation (AMD) of the reaction wheels. Additionally, they assist in maintaining the correct aerobraking drag attitude. They are arranged in couples and are fired in pairs to allow differential firing, so that ideally the resulting net $\Delta V$ is zero [3][4].

| **Thruster** | **Number** | **Thrust** [N] | $I_{\mathbf{sp}}$ [s] | $P_{cc}$ [bar] | **Mass** [kg] | **Propellant** |
|---|---|---|---|---|---|---|
| MR-107N (M) | 6 | 170.0 | 232-229 | 11.2-4.2 | 0.74 | Hydrazine |
| MR-106E (T) | 6 | 22.0 | 235-229 | 12.4-4.5 | 0.52 | Hydrazine |
| MR-103D (A) | 8 | 0.9 | 224-209 | 23.4-5.9 | 0.33 | Hydrazine |

Table 2: Engines Main Characteristics

The masses seen in Table 2 also account for the valve integrated with each system. All the thrusters selected for this mission are flight-proven, with significant heritage from previous space missions. Their extensive operational history provides a solid foundation to meet the propulsive drivers; this knowledge minimizes development risk, enhancing overall mission reliability and predictability, especially during critical mission conditions.

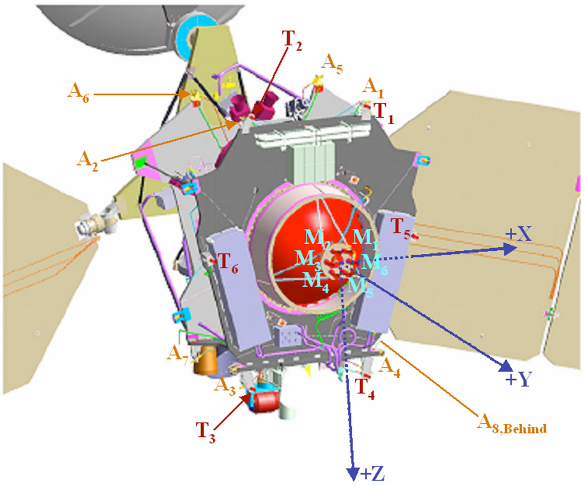


Figure 1: Thrusters and Tank Configuration [8]

### 2.1.2 Propellant

The propulsion design drivers required a system capable of doing frequent start-stop operations, of having a throttleable thrust, and characterized by simplicity, reliability, and predictable behavior. A bi-propellant system ensures a higher performance and better efficiency but involves more complexity in architectures. For these reasons, a monopropellant architecture was selected as the most suitable option [2].

Hydrazine is selected over other possible monopropellants mainly due to its extensive heritage in space applications. Its advantages include excellent long-term stability, ease of decomposition within a catalytic bed, and relatively high specific impulse with respect to other monopropellants. However, employing hydrazine introduces specific risks: it is highly corrosive, meaning that engine plumes or potential leaks could compromise critical spacecraft instruments or components.

### 2.1.3 Pressurant and Feeding Strategy

The pressurant chosen for this mission is helium. It is often utilized in space application because of its low density, its properties, which are consistent with temperature changes, and its non-reactivity with hydrazine. The propulsion system operates in blow-down for most maneuvers, except for the MOI, where a pressure regulator is introduced to guarantee consistent and accurate thruster performance but also to limit the significant fuel consumption of this maneuver. This choice derives from a trade-off between tank dimensions and required precision: a fully regulated system with a larger helium tank, sufficient to maintain constant pressure throughout the entire mission, would have resulted in significant mass and volume penalties. However, relying entirely on a blow-down system would lack the precision and consistency essential for the critical MOI maneuver[3][5].

### 2.1.4 Tanks and Pipes

The MRO spacecraft is equipped with a cylindrical propellant tank designed to contain approximately 1187 kg of hydrazine, approximately 70% of which is allocated to the MOI [2][3]. A separate cylindrical tank, positioned on the side of the main propellant tank, contains high-pressure helium gas that, through a regulator, goes into the propellant tank where it puts the hydrazine under pressure. During the cruise and Mars approach phases, the gradual hydrazine consumption led to a gradual pressure decrease in the tank, dropping from a peak of 205 psi to roughly 190 psi following MOI [8]. From the pressurized tank, hydrazine is directed through metal tubing to each thruster that, by a valve, can be fired independently [5].

## 2.2 Reverse Sizing

The reverse sizing of the MRO propulsion subsystem is carried out based on the total $\Delta V$ required for the mission, reserve excluded, the dry mass at launch and the engine parameters. These data are taken from the literature and summarized in the Table 3 (refer to Table 2 for thrusters data). Some assumptions have also been made to perform the analysis and they will be underlined in the following paragraphs. During the reverse sizing process, the contributions of all three propulsive systems are considered. To this end, the total $\Delta V$ has been divided into three components to consider the different amounts spent by MR-107N, MR-106E and MR-103D engines.

| $\Delta V_{\text{MR107N}}$ | $\Delta V_{\text{MR106E}}$ [m/s] | $\Delta V_{\text{MR103D}}$ [m/s] | $M_{\text{dry}}$ [kg] | $P_{\text{cc, max}}$ [Pa] | $P_{\textbf{cc, min}}$ m/s¿m/s [Pa] |
|---|---|---|---|---|---|
| 1010 | 385 | 150 | 1031 | 2340000 | 420000 |

Table 3: Reverse Sizing Data [3]

- **Propellant Sizing** Firstly, a 20% system-level margin is applied to dry mass at launch. The estimated wet mass can be retrieved using the specific impulse of the engines, the $\Delta V$ and the standard gravitational acceleration constant $g_0 = 9.81\,\text{m/s}^2$ using Tsiolkovsky's equation:

$$M_{\text{wet}} = M_{\text{dry}}\, e^{\frac{\Delta V}{I_{\text{sp}}\, g_0}} \tag{1}$$

  The estimated propellant mass is obtained by subtracting the dry mass from the wet mass. A 5.5% margin is then applied to this value (3% for fuel residuals, 2% for ullage uncertainty, and 0.5% for loading uncertainty). The volume occupied by the propellant can be computed dividing the obtained mass by the density of hydrazine ($\rho_{\text{prop}} = 1010\,\text{kg/m}^3$). An additional 10% margin is then applied to account for possible unusable volume. The procedure described is applied to all three engine groups, considering the $\Delta V$ and specific impulse associated with each. The individual contributions are then summed to obtain the total quantities.

- **Pressurant Sizing** As anticipated in the previous sections, the pressurization system operates in two different ways depending on the mission phases: pressure-regulated and blow-down. The two systems require different reverse engineering approaches, as described below. It is important to note that, due to a lack of information, the initial and final pressures in the gas tank are estimated as follows:

$$P_{\text{gas, f/i}} = P_{\text{cc, min/max}} + \Delta P_{\text{feed}} + \Delta P_{\text{inj}} \tag{2}$$

  where $P_{cc}$ is taken from the engine datasheet, while $\Delta P_{feed}$ and $\Delta P_{inj}$ represent respectively the losses due to the feeding lines (35 kPa) and the injection system (estimated at 30% of the chamber pressure).

*Blow-Down.* The blow-down ratio $B$, computed as the ratio between $P_{\text{gas,i}}$ and $P_{\text{gas,f}}$, allows the calculation of the initial gas volume as follows:

$$V_{\text{gas, bd}} = \frac{V_{\text{prop, bd}}}{B-1} \tag{3}$$

Subsequently, the mass of gas involved in the blow-down system can be computed as:

$$M_{\text{gas, bd}} = \frac{P_{\text{gas, i}} \cdot V_{\text{gas, i}}}{R_{\text{He}} \cdot T_{\text{tank}}} \tag{4}$$

where $R_{\text{He}} = 2077.3\,\text{J}/(\text{kg} \cdot \text{K})$ and the tank temperature is assumed as $T_{\text{tank}} = 293\,\text{K}$. A mass margin of 20% is then applied to the obtained value. This procedure is carried out for both MR-106E and MR-103D engine groups, since these are the systems that primarily rely on a blow-down system.

*Pressure-Regulated.* During MOI, the pressurization system switches to a pressure-regulated system that maintains the tank pressure constant. The mass of helium required for this phase can be calculated assuming an adiabatic transformation:

$$M_{\text{gas, pr}} = \frac{P_{\text{gas, f}} \cdot V_{\text{prop, pr}} \cdot \gamma_{\text{gas}}}{R_{\text{He}} \cdot T_{\text{tank}} \cdot \left(1 + \frac{P_{\text{gas, f}}}{P_{\text{gas, i}}}\right)} \tag{5}$$

where $\gamma_{\text{He}} = 1.67$ and again $T_{\text{tank}} = 293\,\text{K}$. Another 20% mass margin is applied to this value. The fraction of volume occupied by helium can then be computed using the ideal gas law.

In analogy with the propellant sizing, the different contributions are summed to obtain the total values of mass and volume related to the pressurant.

- **Tanks Sizing** As mentioned before, the propellant and gas tanks are assumed to be cylindrical with two hemispherical caps and made from titanium ($\rho_{\text{Ti}} = 2087\,\text{kg/m}^3$, $\sigma_{\text{Ti}} = 950\,\text{MPa}$). With these reasonable hypotheses, it is possible to size two separate tanks for propellant and pressurant. Firstly, a 1% bladder margin is applied to the tank volume. Holding the assumption of $V_{\text{cyl}} = V_{\text{sph}}/3$, the radius of the tank and the height of its cylindrical part can be computed from geometric considerations.

  Subsequently, the thickness of the spherical part and the cylindrical one can also be obtained as:

$$t_{\text{sph}} = \frac{P_{\text{gas, i}} \cdot r_{\text{tank}}}{2 \cdot \sigma} \qquad t_{\text{cyl}} = \frac{P_{\text{gas, i}} \cdot r_{\text{tank}}}{\sigma} \tag{6}$$

  Having the needed quantities, it is now possible to retrieve the masses of the spherical and cylindrical parts of the tanks:

$$M_{\text{sph}} = \rho_{\text{Ti}} \cdot \frac{4}{3}\pi \left( (r_{\text{tank}} + t_{\text{sph, gas}})^3 - r_{\text{tank}}^3 \right) \tag{7}$$

$$M_{\text{cyl}} = \rho_{\text{Ti}} \cdot \pi \left( (r_{\text{tank}} + t_{\text{cyl}})^2 - r_{\text{tank}}^2 \right) \cdot h_{\text{cyl}} \tag{8}$$

  The final mass of the tanks can be finally obtained summing the two contributions.

- **Mass Budget** The total mass of the propulsion subsystem (propellant excluded) can be calculated summing the masses of the thrusters (provided in Table 2), the tanks and the gas.

- **Power Budget** The table below shows the maximum power required by each model considering the valve power, the valve heater power and the catalytic bed power of one thruster only per type.

| | MR-107N | MR-106E | MR-103D |
|---|---|---|---|
| Power [W] | 59.4 | 49.15 | 13.72 |

Table 4: Power Budget

### 2.2.1 Results Analysis

Table 5 shows a comparison of the estimated data with the real ones available from the mission:

| | **w/o Margins** | **Margins** | **w/ Margins** | **Real** |
|---|---|---|---|---|
| $M_{\text{propellant}}$ [kg] | 1017.32 | 5.5 % | 1073.27 | 1149 |
| $M_{\text{pressurant}}$ [kg] | 1.51 | 20 % | 1.81 | - |
| $V_{\text{propellant tank}}$ [m$^3$] | 1.169 | 1 % | 1.181 | 1.175 |
| $M_{\text{propellant tank}}$ [kg] | 17.30 | - | - | - |
| $M_{\text{pressurant tank}}$ [kg] | 3.26 | - | - | - |
| $M_{\text{PS}}$ [kg] | 32.58 | 10 % | 35.83 | - |

Table 5: Estimated Data with and without margins

The propellant mass falls within reasonable margins compared to the actual value. The remaining difference is due to the fact that the total $\Delta V$ considered for the mission includes an additional 135 m/s reserve [3], which were not taken into account in the reverse sizing process. The similarity between the two masses of propellant can be interpreted as evidence that the reverse sizing process can be considered reliable.

Nevertheless, the total mass budget is still likely to be underestimated. With the value obtained, it would account for only 3.5% of the dry mass, which is significantly lower than in most cases. This discrepancy can be explained by the simplified tanks shapes but primarily by the reverse sizing methodology itself, which provides an approximate estimation. In particular, the transition between the blow-down and pressure-regulated systems was modeled in a simplified manner due to the lack of data, which may have led to deviations from reality. Moreover, titanium was assumed as the material for both tanks, but other materials could end up causing significant variations in the final result.

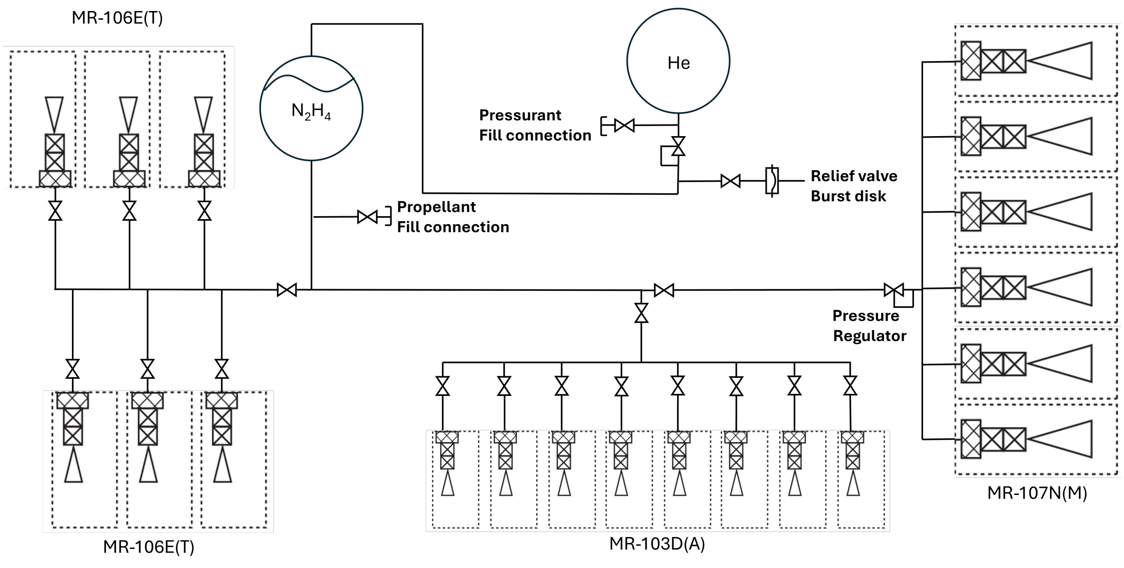


Figure 2: Propulsion architecture schematic

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## Acronyms

**ABM** Aerobraking Maneuver. 1, 2

**ABX** Aerobraking Termination Maneuver. 1

**AMD** Angular Momentum Desaturation. 3

**DLA** Declination of Launch Asymptote. 1

**MOI** Mars Orbit Insertion. 1–6

**MRO** Mars Reconnaissance Orbiter. 1, 3, 5

**OA** Orbit Adjustment. 1, 2

**OTM** Orbit Trim Maneuver. 2

**PSO** Primary Science Orbit. 1

**PSP** Primary Science Phase. 3

**TCM** Trajectory Correction Maneuver. 1–3